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Ten years inside the core systems of one retail network: a Delphi trading platform that stores run on daily, and ASP.NET Core services that keep taking over its functions. Access control for 2,000+ users, fiscal and EDI integrations, and an ongoing Delphi 6 to Delphi 12 client migration.

AT A GLANCE

- 10 years — in production retail systems, 2016 to now
- 2 000+ — users on the access system I built
- 33 — internal systems of the network carry my code
- 2 stacks — Delphi and .NET in parallel, with the business running

WHAT I DELIVERED

- Built UAM, the network's in-house access management system: 2,000+ users on OpenID Connect, ADFS and Kerberos, replacing per-system manual permission handling.
- Leading the trading system client migration from Delphi 6 to Delphi 12/13, with a purpose-built tool that diffs UI and database mappings between versions so the switch does not break store operations.
- Architected and built the network's SMS notification platform: SMPP and GSM API, .NET Web API and MS SQL, designed for peak-volume campaigns.
- Deliver fiscal-law changes end to end across the systems: excise stamps and AED, UKTZED codes, customs simplifications, new VAT rates, all in a live retail network without stopping the stores.
- Built the internal web systems stores work with: journals and audit, recyclables with M.E.Doc integration, a support portal, coupons and reporting.

STACK

Backend: C#, .NET 7 / 6 / 5, .NET Core 3.1, ASP.NET Core, ASP.NET MVC, Web API / REST, SignalR, Worker Services, Hangfire, Quartz, AutoMapper, Flurl.Http

Data: MS SQL Server, T-SQL, EF Core 7, ADO.NET, Oracle, PostgreSQL, Firebird, SSIS, SSRS, SQL CLR

Front end: Angular 13, TypeScript, Kendo UI for Angular, Telerik UI for ASP.NET Core, Razor / MVC views, JavaScript, HTML / CSS, jQuery

Integrations: EDI / ЕДО, M.E.Doc, Oracle EBS, 1C, Kafka, BizTalk, WCF / SOAP, SMPP / GSM API, API Gateway, GemBox.Spreadsheet

Access and security: OpenID Connect, OAuth2, JWT Bearer, ADFS, Kerberos / Negotiate, Active Directory, RBAC

Quality and delivery: xUnit / NUnit, NSubstitute / Moq, AutoFixture, FluentAssertions, SonarQube, Jenkins, Docker, Serilog / NLog, Elastic APM, Git / приватний NuGet

Desktop: Delphi 6 → 12 / 13, Object Pascal, VCL, DevExpress, FastReport, RX Library, Indy, NativeExcel / NativeXml, ADO / OLE DB

Engineering practice: Legacy modernization, Modular monolith, Three-tier architecture, Integration design, Production ownership

Python: Python 3.11, FastAPI, SQLAlchemy 2, Alembic, Pydantic, Typer CLI, Playwright, selectolax, NumPy, pytest, ruff / black, Jinja2

Speech and ML: faster-whisper (CUDA), pyannote.audio, VAD, Speaker diarization, Quality evaluation: RTF, latency, Streaming audio pipelines, WASAPI / sounddevice, PyTorch / GPU

Working with AI agents: Claude Code, MCP-сервери, Custom subagents and hooks, Agent tooling (CDP), Prompt engineering

JS / TS ecosystem: Next.js 16, React 19, NestJS, Prisma, Expo / React Native, Tailwind CSS, Zustand / TanStack Query, MapLibre, Vite, Node.js

KEY PROJECTS

Real-time speech pipeline (ORK) (2026) — Author

Real-time speech recognition from calls: parallel microphone and system-audio capture, VAD-based segmentation, faster-whisper on GPU, pyannote diarization with incremental clustering, translation. A separate regression suite — RTF, latency under load, noise robustness — shows exactly where quality is lost.

Python 3.11, faster-whisper, pyannote.audio, CUDA, FastAPI, NumPy, WASAPI

Trading client migration: Delphi 6 to Delphi 12 (2024 — 2026) — Migration lead

Moving the trading system client to a modern compiler with a core redesign. Built a separate tool that diffs UI and table mappings between the old and new versions, so the migration runs without stopping stores.

Delphi 12/13, Object Pascal, VCL, MS SQL, DevExpress, FastReport

https://github.com/qwert11/d6_d13_dbmapping_checker

Fiscal workflows: excise stamps, AED, customs (2019 — 2026) — Owner of the workstream

End-to-end delivery of legislative change inside the trading system: excise stamp scanning, electronic excise documents, tax invoice adjustments, UKTZED codes, customs simplifications, the 14% VAT rate.

Delphi, T-SQL, MS SQL, EDI, M.E.Doc

UAM — access management (2020 — 2022) — Developer

Centralized permissions and authentication for 2,000+ people in the network: roles and access requests, Active Directory integration, sign-in through OpenID Connect, ADFS and Kerberos.

ASP.NET Core, C#, EF Core, MS SQL, OIDC, ADFS, Angular

Real-estate data pipeline (2026) — Author

A listings parser with pagination, normalization and price history: SQLAlchemy 2 models with Alembic migrations over PostgreSQL, two transports — plain HTTP and Playwright for protected pages — a Typer CLI, pydantic-settings config and pytest coverage.

Python, SQLAlchemy 2, Alembic, PostgreSQL, Playwright, selectolax, Typer, pytest

Legacy system map (2026) — Author

A tool for mapping the trading system before migration: Python parses .dfm and .pas files, extracts stored-procedure calls and screen relations, and a Next.js app renders it as an annotated map.

Python, Next.js, TypeScript, PostgreSQL

EXPERIENCE

Freelance and personal projects — AI / Python and Full Stack Developer

2026-03 — present · Dnipro · remote

- A real-time speech pipeline: faster-whisper on CUDA, pyannote diarization, VAD, WASAPI capture.
- And a regression suite for it: RTF, latency under load, noise robustness, quality ceilings — decisions come from measurements, not impressions.
- A real-estate parser in Python: SQLAlchemy 2 and Alembic over PostgreSQL, Playwright and selectolax, a Typer CLI, pytest and ruff.
- A listings pipeline over six sources with geocoding, deduplication and catalog publishing through GitHub Actions.
- A legacy-code analyzer: Python parses .dfm and .pas files and diffs stored-procedure calls, Next.js renders the system map with annotations.
- Client work: NestJS with Prisma and PostgreSQL, Next.js 16 with React 19, a mobile app on Expo with MapLibre maps.
- Daily work with AI agents: MCP servers, custom subagents and hooks, and my own CDP browser inspector as agent tooling.

ATB-Market — Lead Application Programmer

2016-06 — present · Dnipro

- Own the client side of the trading system that the network's stores run on daily: supply documents, pricing and fiscal invoices, MS SQL.
- Since 2018 also build internal services on ASP.NET Core and Angular: access, audit, reporting, integrations — 33 of the network's systems carry my code.
- UAM: access management for 2,000+ users on OpenID Connect, ADFS and Kerberos.
- Network integrations: EDI and e-document flow, M.E.Doc, Oracle EBS, 1C, Kafka, BizTalk, SSIS and SSRS.
- Fiscal-law changes delivered end to end: excise stamps and AED, UKTZED, customs simplifications, VAT rates, in a live retail network.
- Leading the client migration from Delphi 6 to Delphi 12/13 with an in-house tool that diffs UI and DB mappings between versions.
- Engineering practice: xUnit and NSubstitute tests, SonarQube and Jenkins, Serilog, Docker, a private NuGet feed.

Freelance and private practice — Full Stack Developer

2015-07 — 2016-07 · Kryvyi Rih

- An automated car dealership system on Java and Firebird: vehicles, sales and reporting.

- Online stores on WordPress / WooCommerce and OpenCart, plus a custom WordPress plugin.

Modern Professions Courses — Programming Instructor

2015-01 — 2016-01 · Kryvyi Rih

- A beginners' course: OOP, Java Core, Hibernate, SQL, from first programs to working with a database.
- Wrote the course materials and labs, and took groups from zero to their first projects.

Kryvyi Rih Iron Ore Combine — Engineer Programmer

2013-06 — 2016-06 · Kryvyi Rih

- A system for ore extraction and shipment accounting, developed and maintained in production.
- Automated per-shift distribution of consumables and hazard-pay meals.
- Maintained production-preparation and budgeting systems; implemented Firebird table synchronization.

OSSystem — Developer

2012-11 — 2013-04 · Odesa

- Development on Master INSURANCE, an insurance automation system: three-tier architecture with ORM over Oracle, reports in XL Report.
- Maintained Expo, a system tracking vehicle and container movement across port territories.

DOMAIN KNOWLEDGE

- Excise stamps (eAcciz / AED): Stamp scanning, electronic excise documents, document number in the invoice
- Tax invoices and adjustments: Issuing, adjustments, field rules aligned with the tax service
- UKTZED commodity codes: Commodity code synchronization across the network's systems
- Customs simplifications: Customs declarations, import receipts into warehouses, reporting
- EDI and e-document flow: Orders, waybills, pooled packaging, supplier exchange
- VAT rates and pricing regions: Cross-system rate changes, region setup when a store opens

EDUCATION AND CERTIFICATES

- Computer Engineering · computer systems and networks — Zaporizhzhia Institute of Economics and Information Technologies (2008 — 2011)
- Mining Engineer, underground development — Kryvyi Rih Technical University (1998 — 2003)
- Angular 9 — Luxoft (2021)
- Programming in Delphi — Institute of Modern Professions (2006)

CODE ON GITHUB

- Housing catalog across nine regions — six-source aggregator with a map, price indices and a GitHub Actions build
<https://github.com/qwert11/budynky-zahid>
- ML Formula Wiki — 72 machine-learning formulas from Raschka's book, rendered with KaTeX
<https://github.com/qwert11/ML-Formula-Wiki>
- M3 to M4 Mapping Checker — diffs UI and stored-procedure calls between Delphi 6 and Delphi 13
https://github.com/qwert11/d6_d13_dbmapping_checker
- TeamBeacon — a real-time beacon for finding your group in a crowd: ASP.NET Core, SignalR, Docker
<https://github.com/qwert11/TeamBeacon>
- SQL Profiler Normalizer — turns profiler sp_executesql output into a readable EXEC, fully client-side
<https://github.com/qwert11/sql-profiler-normalizer>
- Angle Tool — measures line angles on charts and screenshots in the browser
<https://github.com/qwert11/slope-angle>
- Resume Engine — this site: YAML to Jinja to HTML, PDF, DOCX and Markdown in two languages
<https://github.com/qwert11/resume2>

LANGUAGES

- Ukrainian — native
- English — technical documentation and written communication; spoken at basic level